

$$\textcircled{3} \text{ area}(S) = a(S) = \int_0^a x^{\frac{1}{n}} dx$$

$$\boxed{\int_a^b x^{\frac{1}{n}} dx}$$

$$\therefore a(S) + a(T) = a^{1+\frac{1}{n}}$$

$$\therefore a(S) = a^{\frac{n+1}{n}} - a(T)$$

$$\textcircled{4} a(T) = \int_0^{a^{\frac{1}{n}}} y^n dy = \frac{1}{n+1} a^{\frac{n+1}{n}}$$

$$\textcircled{5} \int_0^a x^{\frac{1}{n}} dx = a(S) = a^{\frac{n+1}{n}} - \frac{1}{n+1} a^{\frac{n+1}{n}} = \frac{a^{\frac{n+1}{n}}}{\frac{n+1}{n}}$$

$$3. a > 0, b > 0, n \in \mathbb{Z}^+$$

$$\int_a^b x^{\frac{1}{n}} dx$$

$$\text{solve: } \int_a^b x^{\frac{1}{n}} dx = \int_0^b x^{\frac{1}{n}} dx - \int_0^a x^{\frac{1}{n}} dx \quad \text{solve:}$$

$$= \frac{b^{\frac{n+1}{n}} - a^{\frac{n+1}{n}}}{\frac{n+1}{n}}$$

$$1. x^{\frac{1}{n}} \uparrow \text{ on } [0, a], a > 0, n \in \mathbb{Z}^+$$

proof: let $0 \leq x_1 < x_2$,

$$\text{if } x^{\frac{1}{n}} > x_2^{\frac{1}{n}}$$

$$\therefore x^n \uparrow \text{ on } [0, a],$$

$$\therefore (x_1^{\frac{1}{n}})^n > (x_2^{\frac{1}{n}})^n$$

$$\therefore x_1 > x_2 \quad \text{contradiction}$$

$$\therefore x^{\frac{1}{n}} < x_2^{\frac{1}{n}}$$

$$2. \int_0^a x^{\frac{1}{n}} dx, a > 0, n \in \mathbb{Z}^+$$

$$\text{solve: } \int_a^b x^{\frac{1}{n}} dx = \int_0^b x^{\frac{1}{n}} dx - \int_0^a x^{\frac{1}{n}} dx \quad \text{solve: } \textcircled{1} \therefore x^{\frac{1}{n}} \uparrow \text{ and bounded on } [0, a],$$

$$\therefore \int_0^a x^{\frac{1}{n}} dx \exists.$$

$\textcircled{2}$

